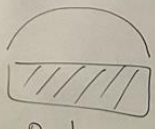
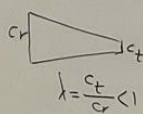


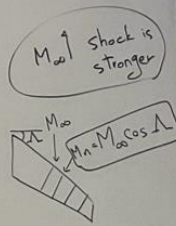
Compressible flow where $\rho \neq \text{cte}$
occurs when $M = \frac{u}{a} > 0.3$



Rectangular
(heavy)



$k = \frac{c_t}{c_r} < 1$



$M_0 \uparrow$ shock is stronger
 $M_1 = M_0 \cos \Delta$

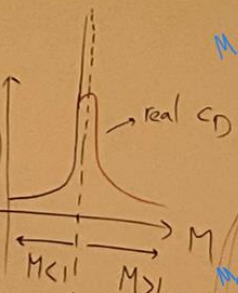
$\rho_\infty = \frac{P_\infty}{RT_\infty}$

$V_\infty^2 = M_\infty^2 \gamma R T_\infty$

$D = C_D \frac{1}{2} \rho_\infty V_\infty^2 S$

$D = C_D \frac{\gamma}{2} P_\infty M_\infty^2 S$

$L = C_L \frac{\gamma}{2} P_\infty M_\infty^2 S$



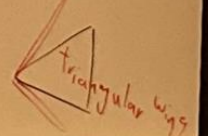
real C_D

$M < 1$ $M > 1$

Sound barrier

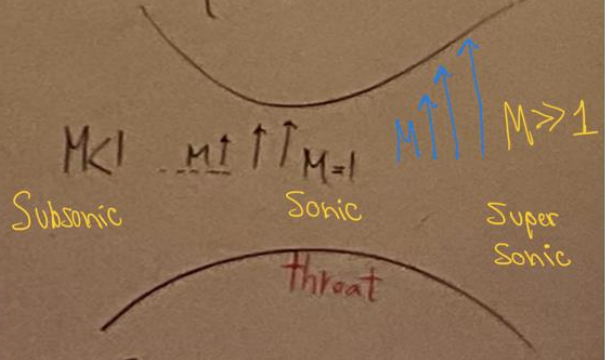


$M_0 \uparrow$ $M_1 \uparrow$
Diamond shape



$M_0 \uparrow$ $M_1 \uparrow$
Triangular wings

Subsonic $M \uparrow$ as $A \downarrow$
Supersonic $M \uparrow$ as $A \uparrow$

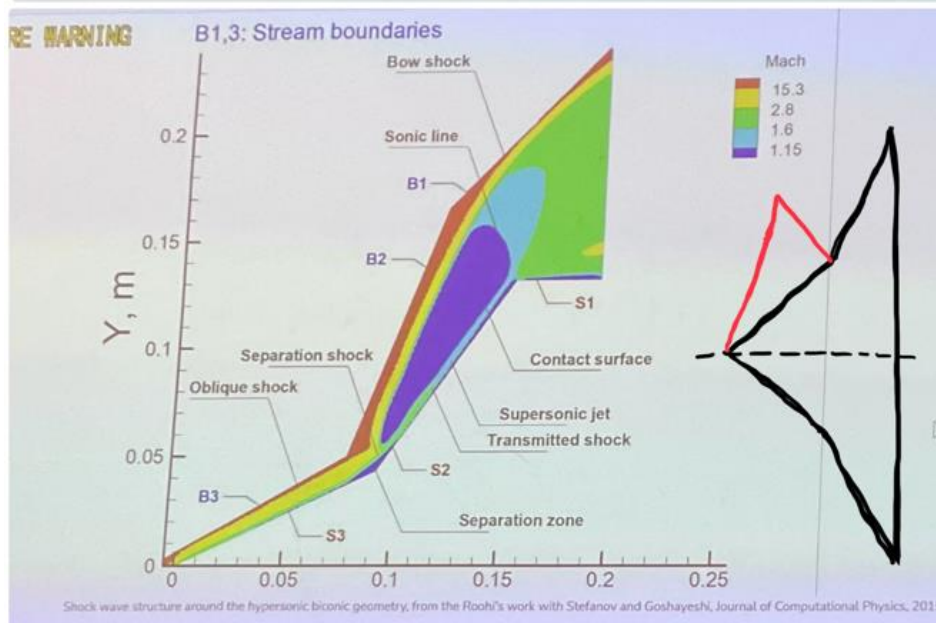
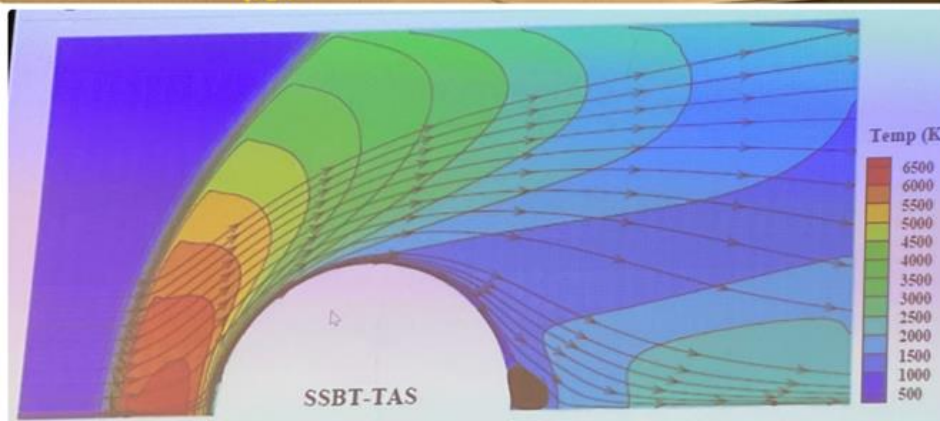
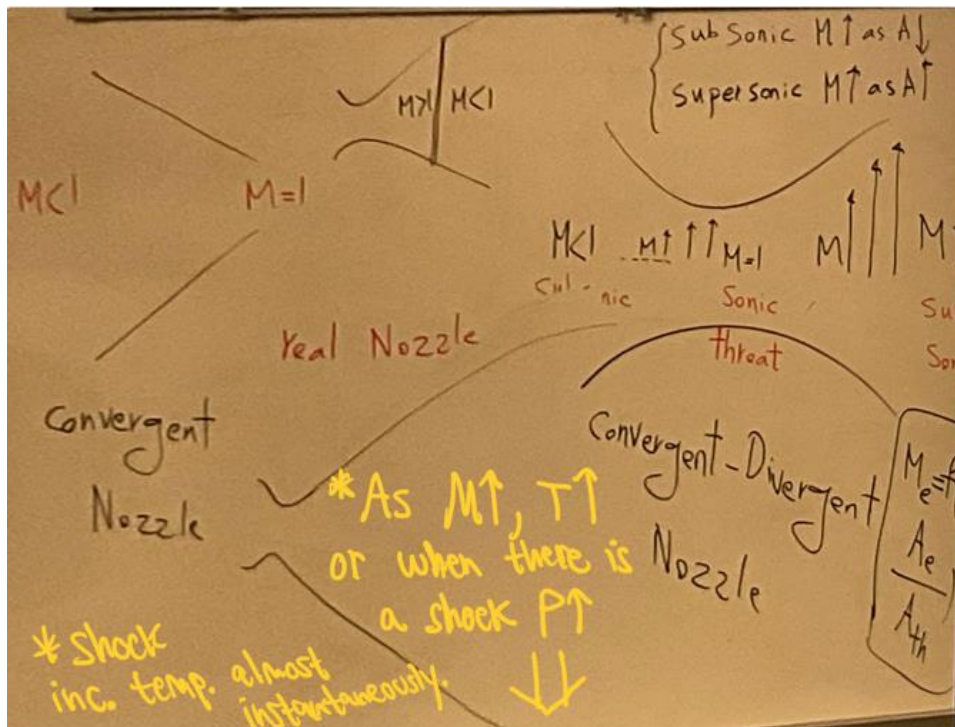


$M < 1$ $M \uparrow \uparrow M = 1$ $M \gg 1$
Subsonic Sonic Super Sonic

Throat

Convergent-Divergent
Nozzle

$M_{\text{exit}} = f\left(\frac{A_e}{A_{th}}\right)$



(SEE NOTES POSTED)